Cancer Genomics: Concepts and Key Genetic Markers

Course Instructor: Chioma Onyido

Learning Objectives

- Understand key concepts and terms in cancer genomics.
- Differentiate between types of cancer-related genes (oncogenes, tumor suppressor genes).
- Identify key genetic markers associated with common cancers.
- Appreciate the clinical importance of genetic testing in cancer.

What is Cancer Genomics?

- **Cancer** is a group of diseases in which abnormal cells in the body grow uncontrollably.

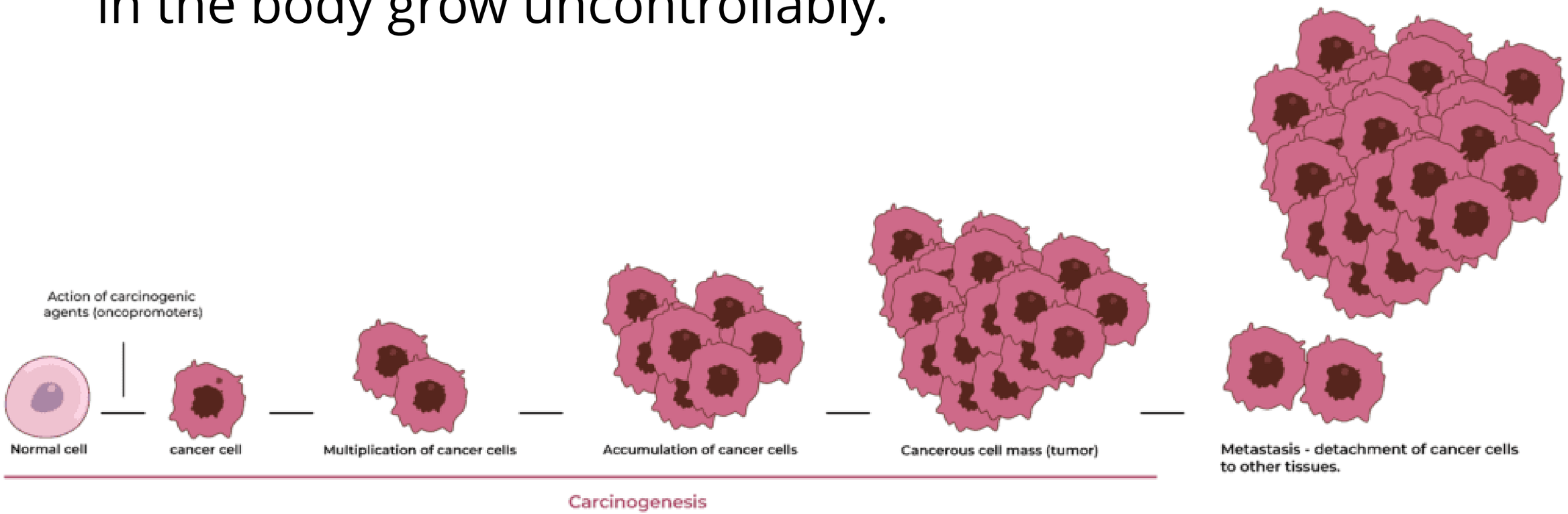

Figure 1: Carcinogenesis (Santos, 2023).

What is Cancer Genomics?

- **Cancer genomics** studies how changes (mutations) in the DNA/ genes contribute to cancer.
- Some mutations are inherited (**germline**), others are acquired (**somatic**).

Types of Cancer-related Genes

Gene Type	Function	Role in Cancer
Oncogenes	Promote cell growth	Mutated oncogenes = uncontrolled growth
Tumor Suppressor	Regulate cell cycle, trigger cell death	Mutated = cells ignore death signals
DNA Repair Genes	Fix DNA damage	Mutated = more mutations accumulate

Key Concepts & Definitions

- **Mutation**: A permanent change in the DNA sequence that may or may not affect cell function.
- **Driver Mutation**: Directly contributes to cancer development.
- **Passenger Mutation**: Occurs alongside but doesn't cause cancer.

Key Concepts & Definitions

- **Genomic Instability:** High mutation rate in cancer cells.
- **Loss of Heterozygosity (LOH):** Loss of one allele of a gene (usually tumor suppressor).
- **Penetrance**: likelihood that a mutation leads to disease.
- **Expressivity**: how much the trait/disease is expressed

Key Concepts & Definitions

- **Tumor heterogeneity**: diversity of mutations within the same tumor.
- **Clonal evolution:** tumor develops as different cells acquire unique mutations.
- **Microsatellite instability (MSI):** common in mismatch repair defects.

Germline vs. Somatic Mutations

- **Germline:** Inherited, found in every cell, passed to offspring.
- **Somatic:** Acquired during life, only in cancer cells, not inherited

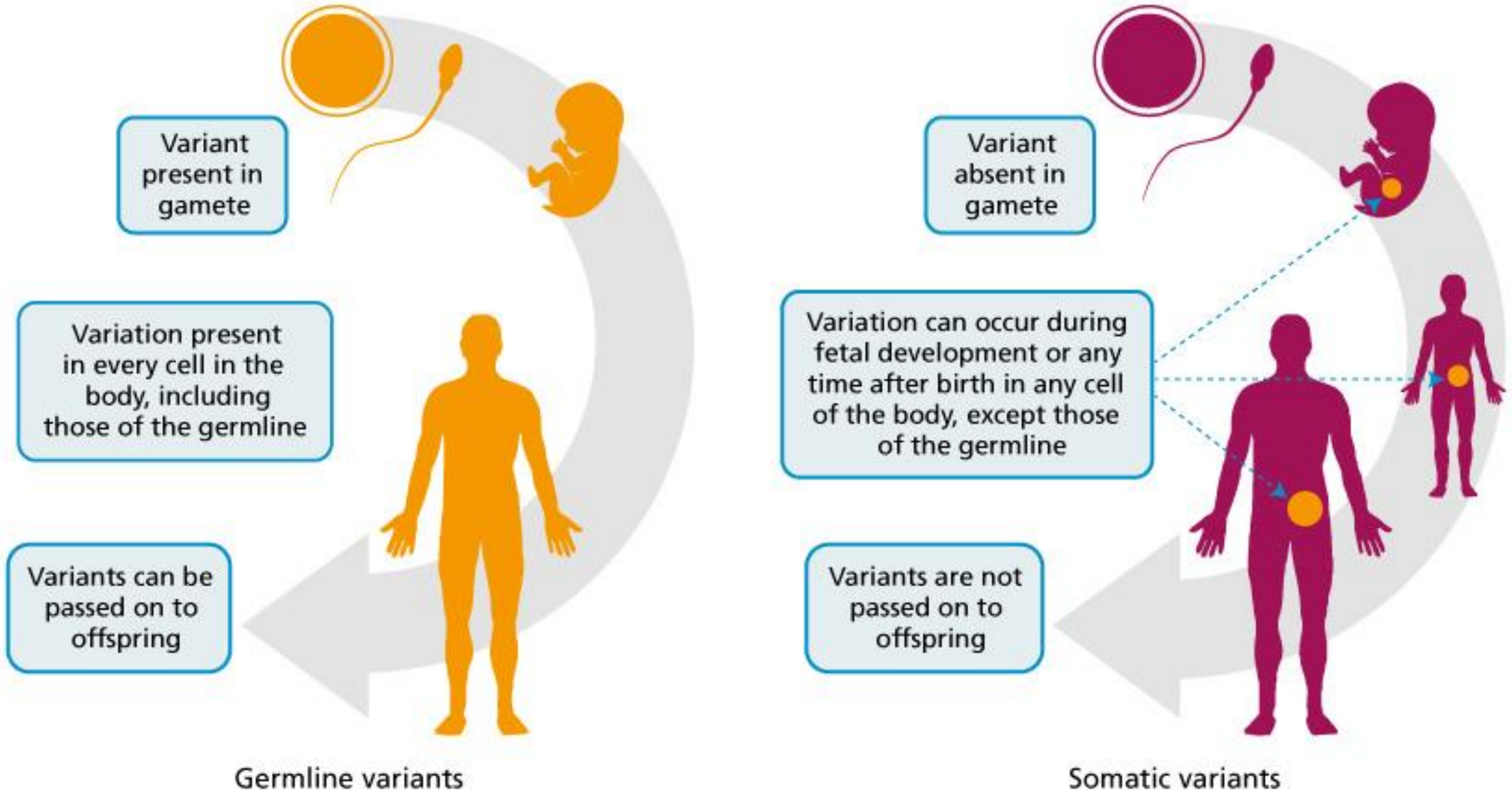

Figure 2: Germline vs Somatic mutations (Genomics Education Program, 2019).

Functional Consequences of Cancer Mutations

- Uncontrolled cell growth.
- Resistance to cell death.
- Inducing angiogenesis (new blood vessels).
- Immune evasion.
- Metastasis and invasion

Key Genetic Markers in Cancer

Marker/Gene	Cancer Type	Role
BRCA1/2	Breast, Ovarian	DNA repair; mutations increase risk
TP53	Many types	Tumor suppressor, common mutation
KRAS	Colorectal, Pancreatic	Oncogene; promotes growth
EGFR	Lung	Growth signaling; target for drugs
HER2	Breast	Receptor protein; target for therapy

Example: BRCA1 and BRCA2

- Involved in **fixing DNA** double-strand breaks.
- Mutations increase **breast** and **ovarian cancer risk** significantly.
- Inherited in an **autosomal dominant pattern**.
- **Targeted therapies** (e.g., PARP inhibitors) are used when BRCA mutation is known.

How Genetic Markers Inform Treatment

- Identify **high-risk** individuals.
- Guide **targeted therapies** (e.g., EGFR inhibitors).
- **Predict** treatment response and resistance.
- **Monitor** disease progression or recurrence

Genetic Testing in Cancer Care

- Performed on blood or tumor samples.
- May include gene panels, whole exome/ genome sequencing.
- Helps **personalize medicine** (precision oncology).
- Requires **genetic counseling** for interpretation.

Practice Questions

1. What is the difference between an oncogene and a tumor suppressor gene?
2. Why are BRCA1 and BRCA2 mutations important in breast cancer?
3. What type of mutation is passed from parent to child?
4. Name one way genetic testing helps in cancer treatment.

Conclusion

- Cancer arises from both **inherited** and **acquired** genetic alterations that disrupt normal cell regulation. Understanding **cancer genomics** gives you the **foundation** to interpret real data.

Coming up next…

Understanding Bioinformatics File Formats:

- ✓ FASTA
- ✓ VCF
- ✓ BAM

Further Reading/ References

- Genomics Education Program (2019). Cancer Genomics. Accessed from https://www.genomicseducation.hee.nhs.uk/cancer-genomics/
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- Hanahan, D. and Weinberg, R.A. (2011). Hallmarks of cancer: the next generation. *Cell*. 144 (5): 646-74. doi:10.1016/j.cell.2011.02.013